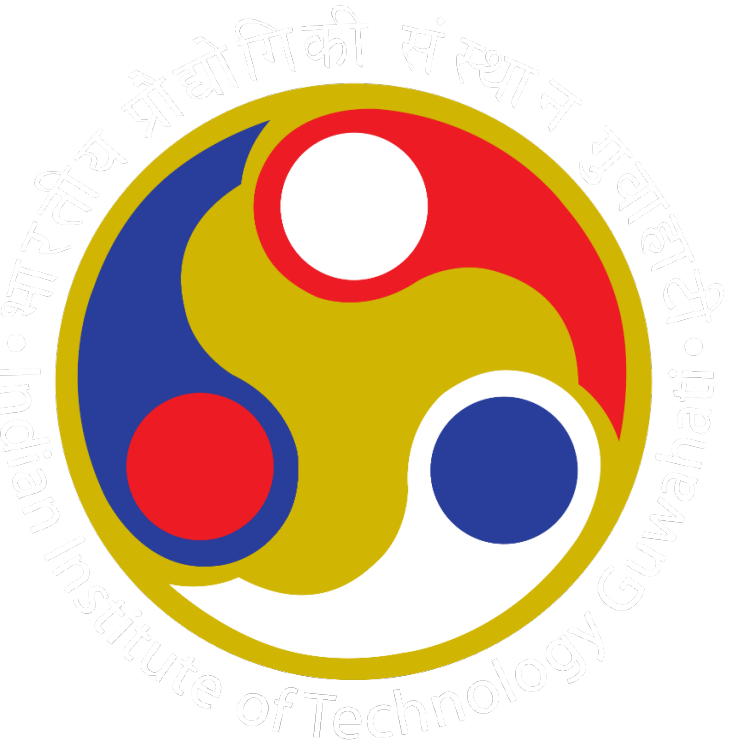
EE517 Analog VLSI Lab

Endsem Project

Design of 2-stage Op-Amp using gm/id Methodology and Layout using Common Centroid Method



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1 Objective

The objective of this project is to design and analyze a two-stage operational amplifier using gm/id methodology and implement its layout using the common centroid technique. The design aims to achieve high gain-bandwidth product (GBW) with low power consumption while satisfying the following specifications:

General Specifications

- Supply Voltage (V_{DD}): 1.8 V
- Slew Rate: ≥ 5 V/μs
- Phase Margin: $\geq 60^\circ$
- Load Capacitance (C_L): 1 pF

Performance Targets

- Gain: ≥ 50 dB
- Common Mode Gain: ≤ -10 dB
- Gain Bandwidth Product (GBW): ≥ 150 MHz
- Output Voltage Swing (Differential Peak-to-Peak): ≥ 800 mV$_{pp}$
- Power Consumption: ≤ 1 mW

After Layout we perform Parasitic extraction and then we compare pre layout simulation and post layout simulation.

2 Theory

The performance of a two-stage operational amplifier is primarily determined by key parameters such as gain, gain-bandwidth product (GBW), phase margin, input/output common-mode range, power supply rejection ratio (PSRR), and slew rate. These parameters dictate the stability, speed, accuracy, and robustness of the amplifier in practical applications.

In analog integrated circuit design, schematic-level simulations provide an ideal representation of circuit performance by considering only the intended devices and components. However, during physical implementation of the circuit in layout, several additional effects are introduced due to fabrication constraints, interconnect routing, and device placement. These effects are collectively referred to as layout-dependent parasitic effects.

Layout design converts the schematic representation of a circuit into a geometrical structure consisting of diffusion regions, polysilicon gates, metal interconnect layers, contacts, and vias that can be fabricated using a specific CMOS technology process (TSMC 180 nm in this experiment). During this transformation, the electrical behavior of the circuit is influenced by unavoidable parasitic elements associated with the physical structure.

2.1 Gain

The voltage gain of an operational amplifier represents its ability to amplify small differential input signals. A high DC gain is essential for accurate amplification and ensures minimal steady-state error in feedback systems. In analog circuits such as amplifiers and filters, insufficient gain leads to poor signal fidelity and reduced precision. For a two-stage op-amp, gain is typically achieved through cascading stages, where the first stage provides high gain and the second stage boosts output swing.

2.2 Gain-Bandwidth Product (GBW)

The gain-bandwidth product defines the trade-off between gain and frequency response. It is a critical parameter that determines how fast the amplifier can operate while maintaining adequate gain. A high GBW is necessary for high-speed applications such as communication systems and data converters. In feedback systems, GBW directly impacts the closed-loop bandwidth, making it a key design constraint for achieving both speed and accuracy.

2.3 Phase Margin

Phase margin is a measure of the stability of the amplifier in a feedback configuration. It indicates how close the system is to oscillation. A phase margin of at least 60° is generally preferred to ensure stable operation with minimal overshoot and ringing. Poor phase margin can lead to oscillations or excessive transient response, making the amplifier unreliable in practical circuits.

2.4 Input Common Mode Range (ICMR)

ICMR defines the range of input voltages over which the amplifier operates correctly while maintaining linearity. It is especially important in differential amplifiers, as exceeding this range causes distortion or improper biasing of transistors. A wide ICMR improves the versatility of the op-amp, allowing it to handle a broader range of input signals.

2.5 Output Common Mode Range (OCMR)

OCMR specifies the allowable range of output voltage swing while maintaining proper operation. It determines how close the output can approach the supply rails. A wider OCMR is crucial for maximizing dynamic range and ensuring efficient utilization of the supply voltage, particularly in low-voltage designs.

2.6 Power Supply Rejection Ratio (PSRR)

PSRR measures the ability of the op-amp to reject variations in the power supply voltage. A high PSRR is essential for maintaining output signal integrity in the presence of supply noise and fluctuations. This is particularly important in mixed-signal and low-noise analog systems where power supply disturbances can degrade performance.

2.7 Slew Rate

Slew rate defines the maximum rate of change of the output voltage with respect to time. It limits the large-signal response of the amplifier. A higher slew rate is required for accurately reproducing fast-changing signals without distortion. If the slew rate is insufficient, the output signal becomes distorted, especially for high-frequency or large-amplitude inputs.

2.8 GM/ID Methodology

The g_m/I_D methodology is a systematic and physics-based design approach widely used in analog CMOS circuit design. It provides a unified framework to design MOS transistors across all regions of operation (weak, moderate, and strong inversion) by relating the transconductance (g_m) to the drain current (I_D). This methodology is particularly useful in deep submicron technologies where traditional square-law models are no longer accurate.

2.8.1 Definition and Physical Meaning

The ratio $\frac{g_m}{I_D}$ is defined as:

$$\frac{g_m}{I_D} \tag{1}$$

This parameter represents the transconductance efficiency of a MOS transistor, i.e., how effectively the device converts bias current into transconductance. A higher value of $\frac{g_m}{I_D}$ indicates higher efficiency, which is desirable for low-power designs.

Physically, it can also be interpreted as the slope of the logarithmic drain current with respect to gate voltage:

$$\frac{g_m}{I_D} = \frac{\partial \ln(I_D)}{\partial V_{GS}} \tag{2}$$

2.8.2 Operating Regions Based on g_m/I_D

The value of $\frac{g_m}{I_D}$ determines the inversion region of operation:

- **Weak Inversion (Subthreshold Region):**
  The drain current exhibits exponential behavior:

  $$I_D \propto e^{\frac{V_{GS}}{nV_T}}$$

  leading to:

  $$\frac{g_m}{I_D} \approx \frac{1}{nV_T}$$

  This region offers very high transconductance efficiency but suffers from low speed due to small current levels.

- **Strong Inversion:**
  The transistor follows a square-law behavior:

  $$I_D \propto (V_{GS} - V_{TH})^2$$

giving:

$$\frac{g_m}{I_D} = \frac{2}{V_{ov}}$$

where $V_{ov} = V_{GS} - V_{TH}$ is the overdrive voltage. This region provides high speed but lower efficiency.

- **Moderate Inversion:**
  This region lies between weak and strong inversion and provides an optimal trade-off between power efficiency and speed. Most analog designs, including operational amplifiers, are biased in this region.

2.8.3 Design Trade-offs Using g_m/I_D

The g_m/I_D parameter directly influences key performance metrics:

- **Gain:** The intrinsic gain of a transistor is given by:

  $$A_v = g_m r_o$$

  Higher g_m/I_D generally leads to higher gain due to improved transconductance efficiency.

- **Bandwidth (GBW):**

  $$GBW \approx \frac{g_m}{2\pi C_L}$$

  Increasing g_m improves bandwidth, but this requires higher current if g_m/I_D is low.

- **Power Consumption:**

  $$P = V_{DD} \cdot I_D$$

  Lower current operation (higher g_m/I_D) reduces power consumption.

- **Slew Rate:**

  $$SR = \frac{I_D}{C_C}$$

  Higher bias current improves slew rate, creating a trade-off with power.

Thus, selecting an appropriate g_m/I_D value is crucial for balancing gain, speed, and power consumption.

2.8.4 Design Procedure

The g_m/I_D methodology follows a structured design flow:

1. Define design specifications such as gain, GBW, slew rate, and power.
2. Determine required transconductance from GBW:

   $$g_m = 2\pi \cdot GBW \cdot C_L$$

3. Select a suitable $\frac{g_m}{I_D}$ value based on desired trade-offs.

4. Compute the drain current:
$$I_D = \frac{g_m}{(g_m/I_D)}$$

5. Use pre-characterized g_m/I_D curves (obtained from simulation) to extract:
   - Transistor dimensions (W/L)
   - Gate voltage (V_{GS})
   - Output resistance (r_o)

6. Validate performance through simulation and iterate as needed.

2.8.5 Advantages of GM/ID Methodology

- Provides a unified approach across all inversion regions
- Enables efficient power-performance optimization
- Reduces dependency on inaccurate analytical models
- Directly links device-level parameters to circuit-level specifications
- Highly suitable for modern low-voltage, low-power analog design

2.8.6 Relevance to Two-Stage Op-Amp Design

In a two-stage operational amplifier:

- The input differential pair is typically designed with higher g_m/I_D for better gain and noise performance.
- The second stage is biased at lower g_m/I_D (higher current) to achieve higher slew rate and output drive capability.
- The methodology allows systematic sizing of transistors to meet specifications such as GBW, phase margin, and power consumption.

Thus, the g_m/I_D methodology provides a powerful and intuitive framework for designing high-performance analog circuits such as operational amplifiers.

2.9 Common Centroid Layout Technique

In analog integrated circuit design, device matching plays a crucial role in determining circuit performance. Mismatch between transistors or passive components can degrade parameters such as gain, offset voltage, linearity, and common-mode rejection ratio (CMRR). The common centroid layout technique is a widely used method to minimize mismatch caused by process variations and layout-induced gradients.

2.9.1 Sources of Mismatch

Mismatch in integrated circuits arises due to two primary factors:

- **Random Variations:** Caused by fluctuations in dopant concentration, oxide thickness, and lithographic imperfections. These are statistical in nature and cannot be completely eliminated.
- **Systematic Variations:** Caused by process gradients such as variations in temperature, oxide thickness, or stress across the chip. These variations are spatially dependent and can be minimized through careful layout techniques.

The common centroid technique primarily targets the reduction of systematic mismatch.

2.9.2 Concept of Common Centroid

The basic idea of the common centroid layout is to arrange identical devices such that their geometric centers coincide. This ensures that all devices experience nearly identical process gradients, thereby averaging out systematic variations.

For example, consider two matched devices A and B. Instead of placing them side by side as:

$$A \quad B$$

they are split into smaller segments and interdigitated in a symmetric pattern such as:

$$A\ B\ B\ A$$

or more generally:

$$A\ B\ B\ A \quad / \quad B\ A\ A\ B$$

This arrangement ensures that the centroid of device A coincides with that of device B, effectively cancelling linear process gradients.

2.9.3 Interdigitation and Segmentation

To implement common centroid layout, large devices are divided into multiple smaller unit devices (fingers). These unit devices are then interleaved (interdigitated) in a symmetric pattern.

- **Segmentation:** Improves matching by averaging local variations.
- **Interdigitation:** Ensures that each device samples the same environment.

For example, a 1:1 matched pair can be implemented as:

$$A_1\ B_1\ B_2\ A_2$$

For ratioed devices (e.g., 2:1), patterns such as:

$$A\ B\ A\ B\ A\ B$$

are used to maintain symmetry.

2.9.4 Gradient Cancellation

Assume a linear gradient exists across the layout (e.g., threshold voltage variation). In a simple side-by-side layout, one device experiences a higher value than the other, leading to mismatch.

In a common centroid arrangement, each device is distributed symmetrically across the gradient. As a result, the average value seen by both devices becomes equal, thereby cancelling first-order gradient effects.

2.9.5 Layout Considerations

While implementing common centroid layouts, the following guidelines must be followed:

- Maintain strict symmetry about both horizontal and vertical axes
- Use identical orientation for all unit devices
- Ensure equal routing parasitics for matched devices
- Place dummy devices at the edges to minimize edge effects
- Keep interconnect lengths and surroundings identical

2.9.6 Advantages

- Significant reduction in systematic mismatch
- Improved matching of critical device pairs
- Enhanced performance in differential circuits
- Better control over offset and gain accuracy

2.9.7 Application in Two-Stage Op-Amp

In a two-stage operational amplifier, the common centroid technique is essential for:

- **Differential Pair:** Ensures accurate matching, reducing input offset voltage.
- **Current Mirrors:** Maintains precise current ratios, improving bias stability.
- **Active Loads:** Enhances gain and symmetry in operation.

Poor matching in these components can lead to degraded gain, increased offset, and reduced common-mode rejection. Therefore, the use of common centroid layout is critical for achieving high-performance analog designs.

3 Pre parasitics extraction results

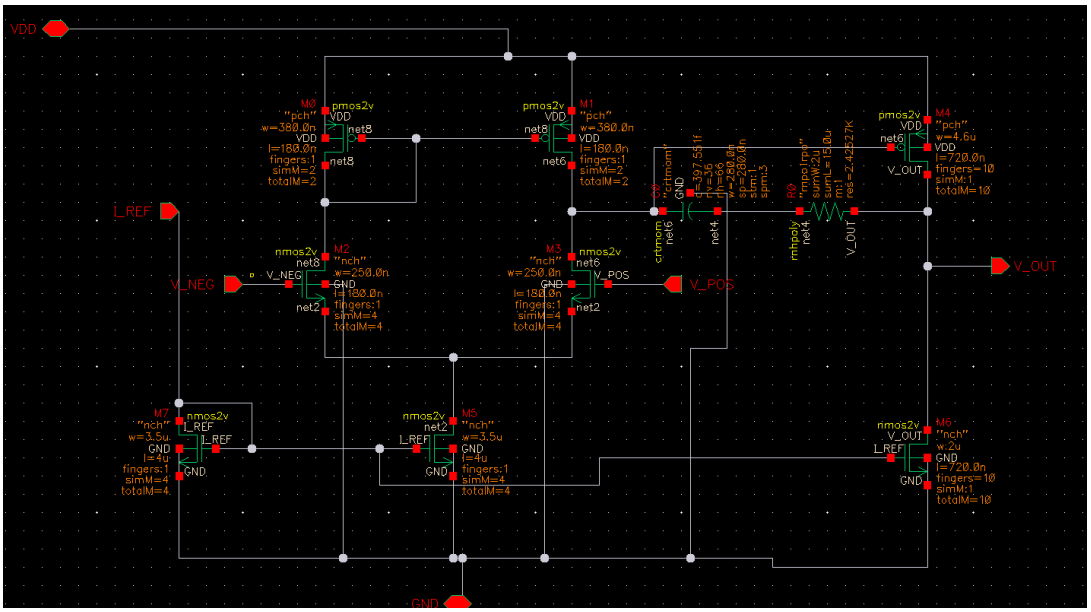


Figure 1: 2 Stage Op amp design

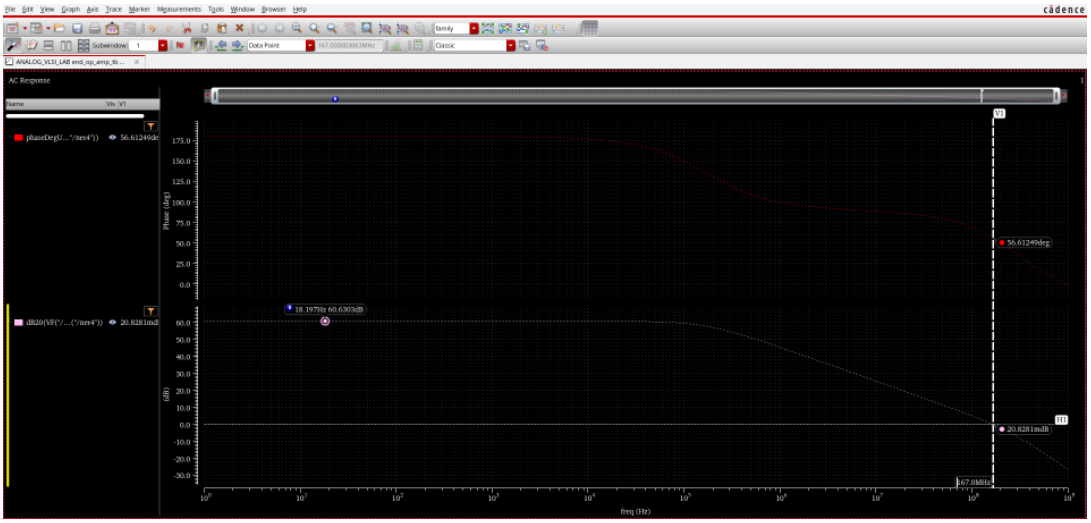


Figure 2: Gain GBW Phase

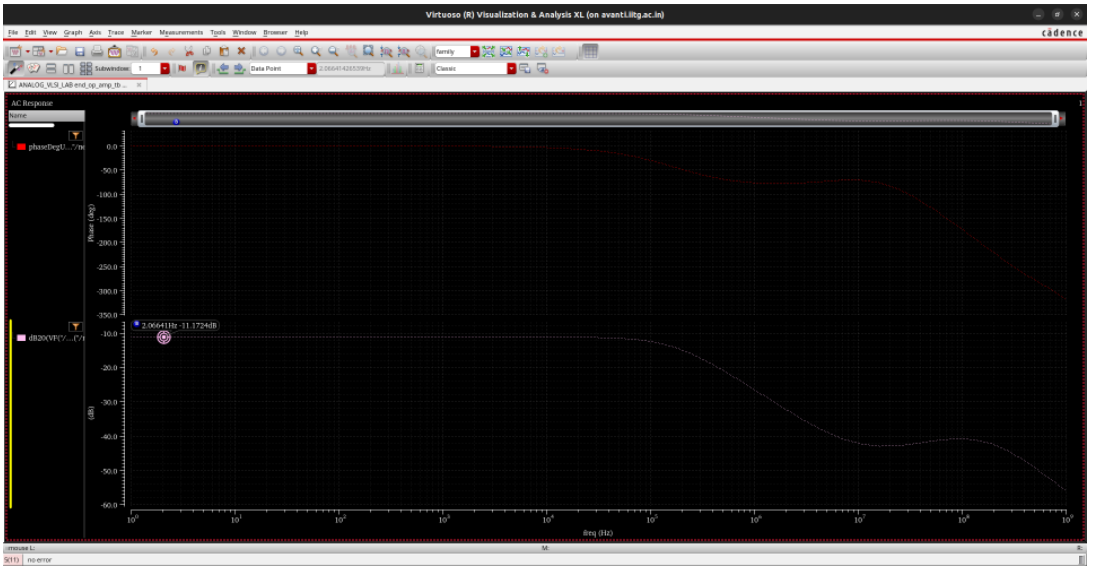


Figure 3: Common Mode Gain

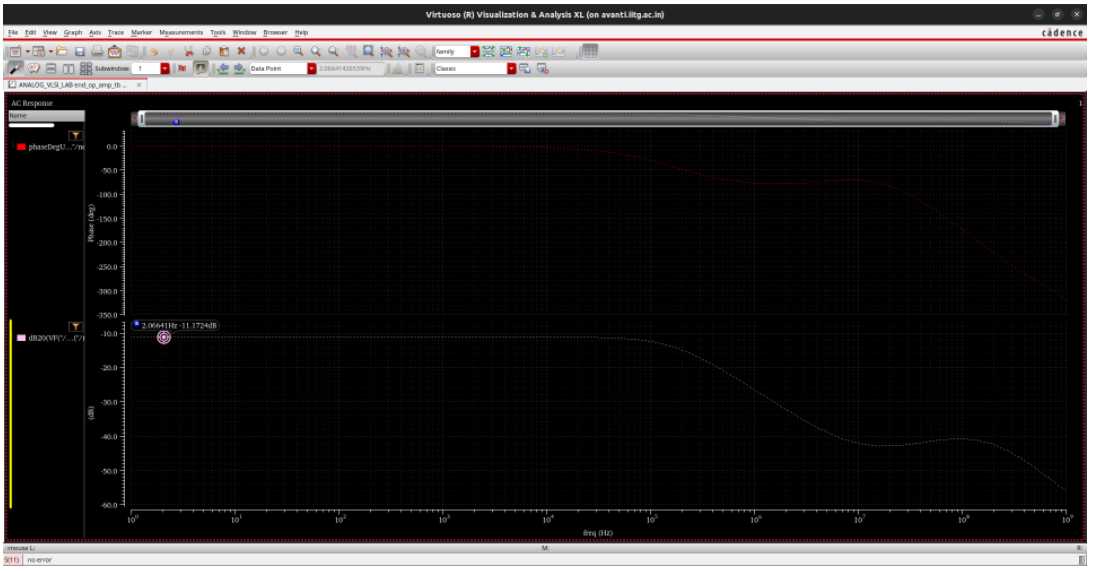


Figure 4: Common Mode Gain

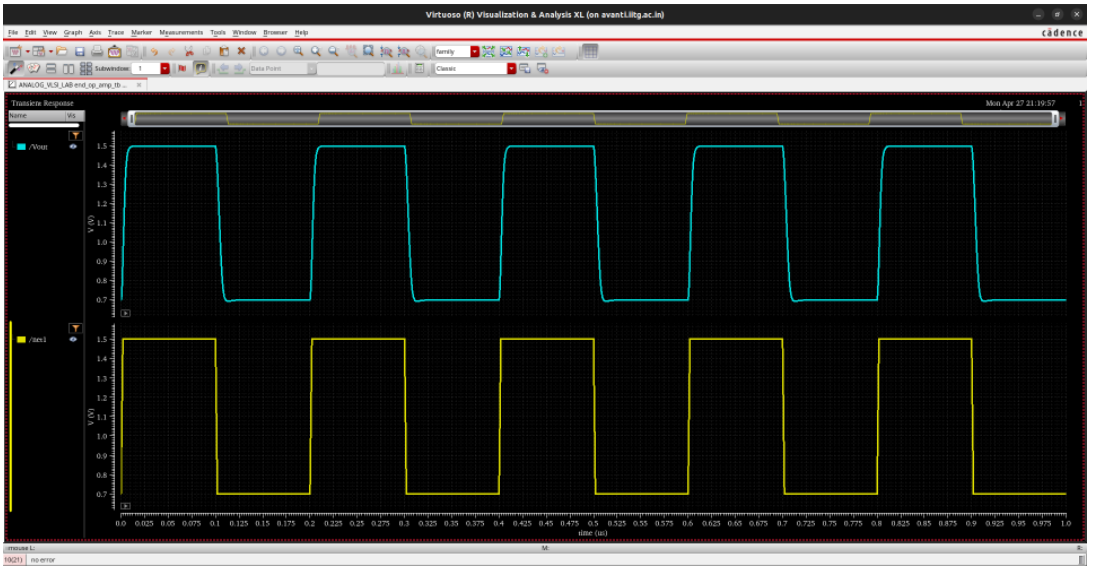


Figure 5: Pulse input(5Mhz) and output

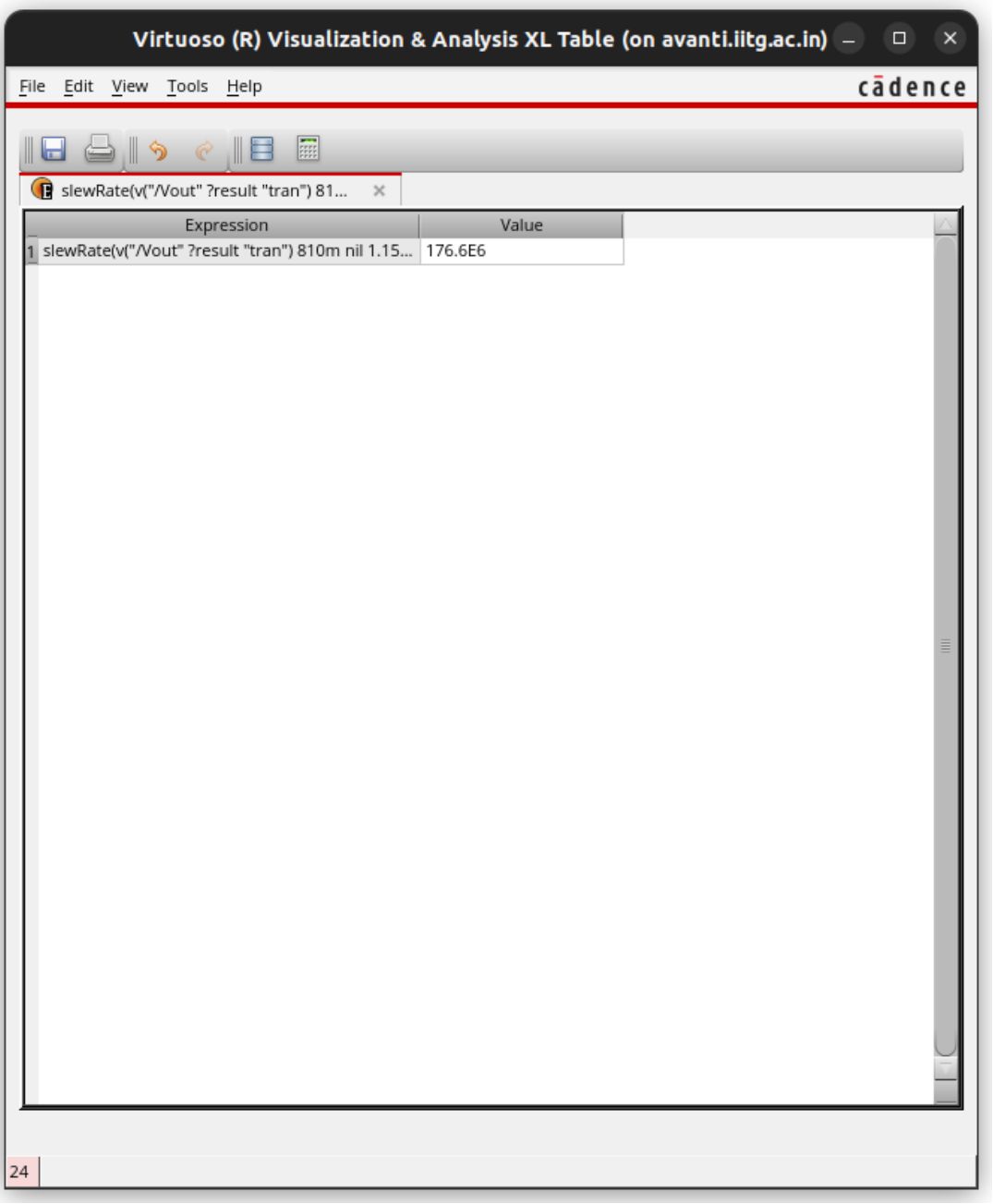


Figure 6: Slew Rate

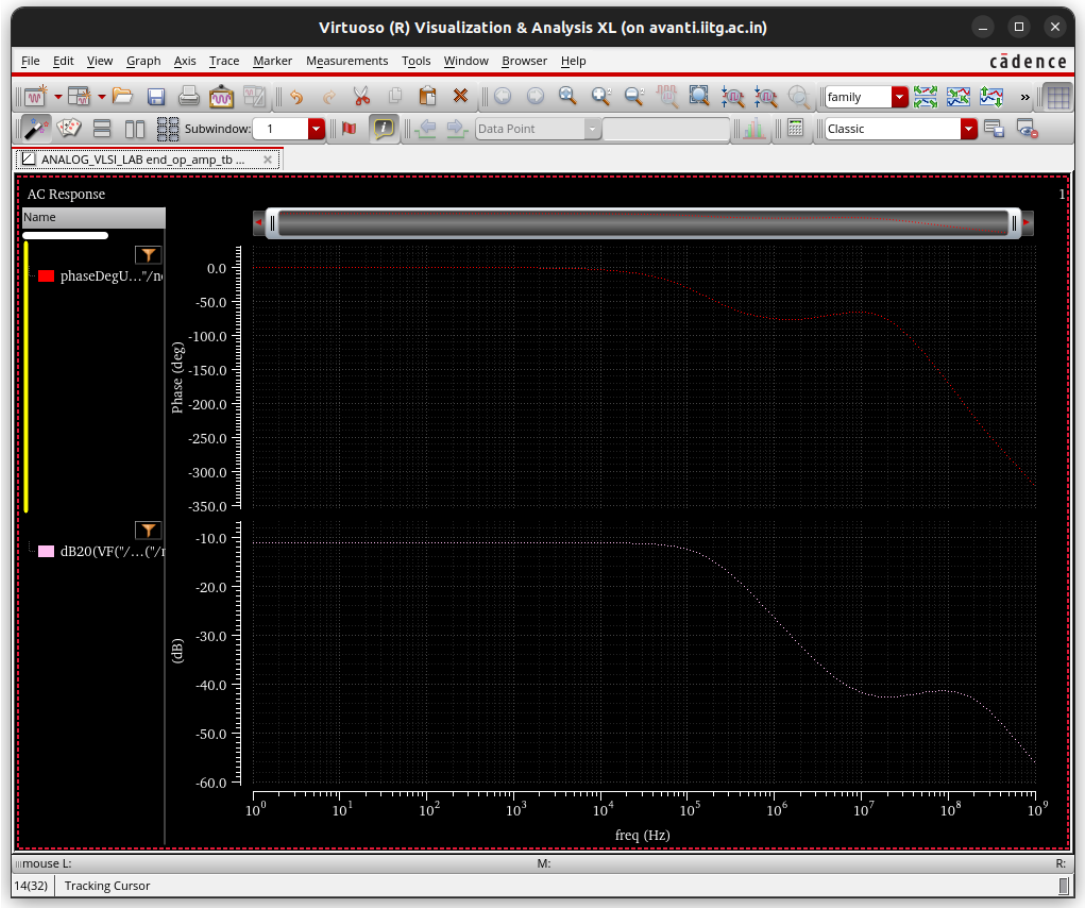


Figure 7: PSRR

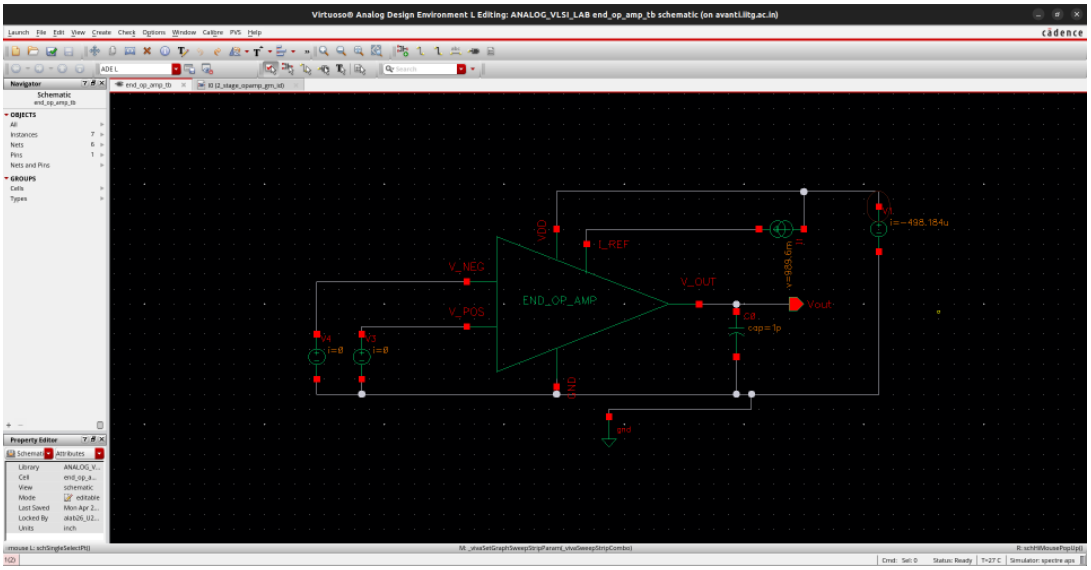


Figure 8: Power Consumed

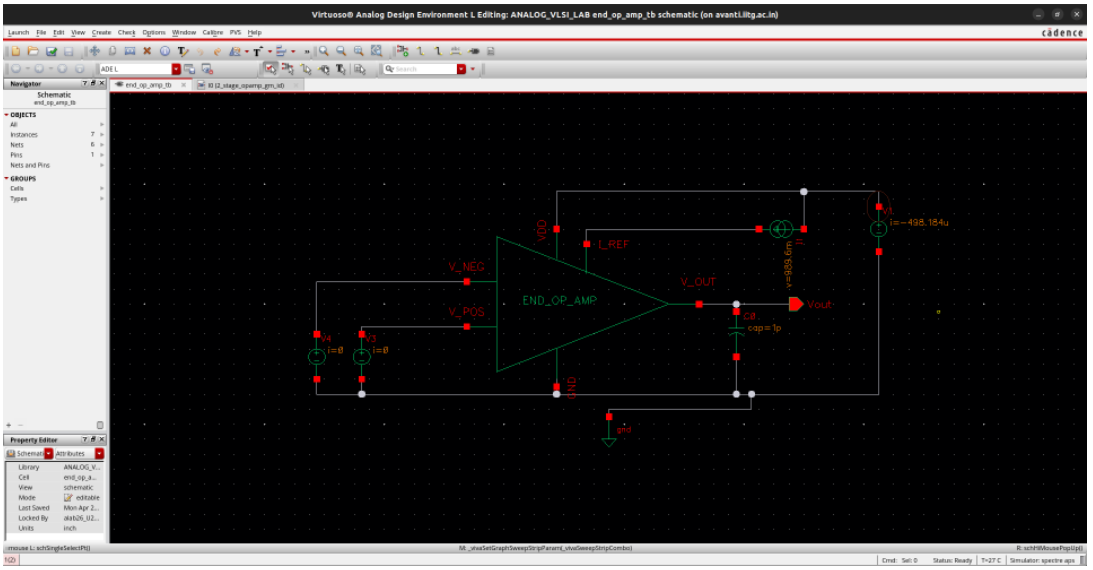


Figure 9: OCMR

4 Post parasitics extraction results

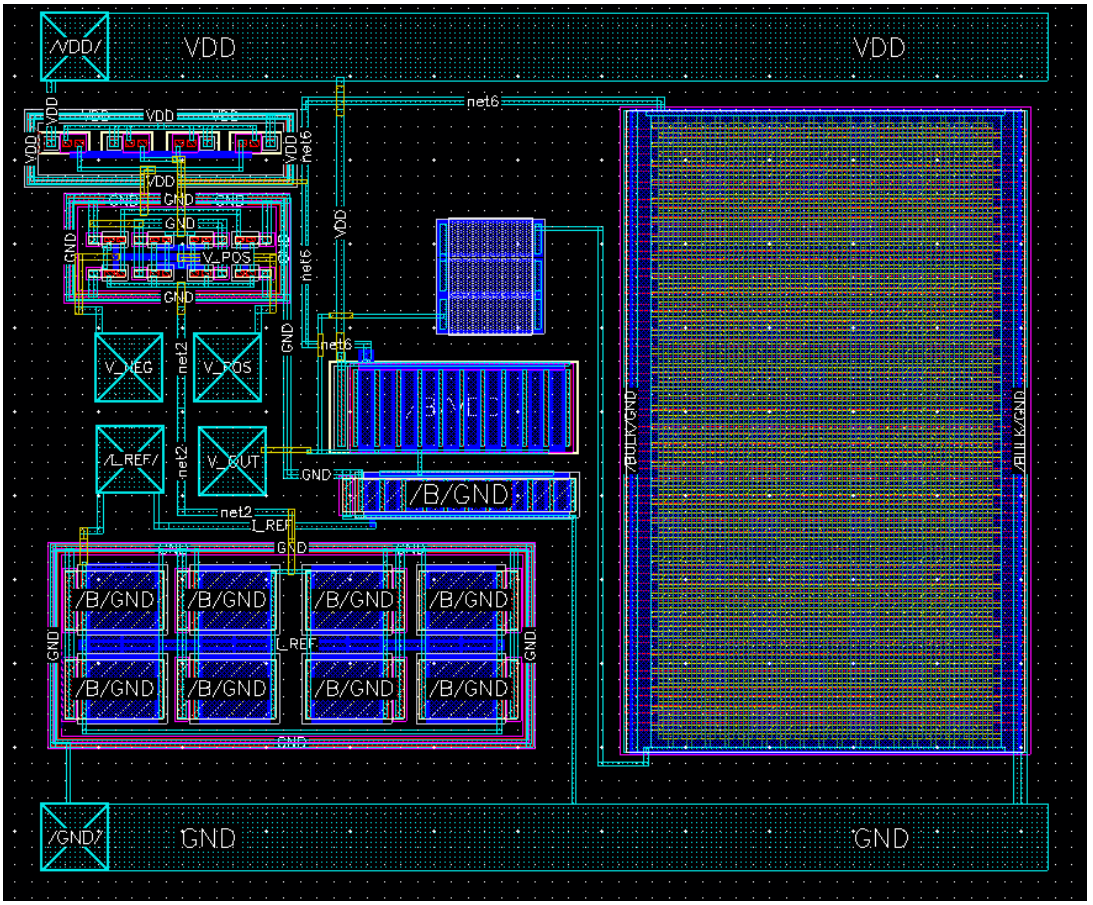


Figure 10: Layout

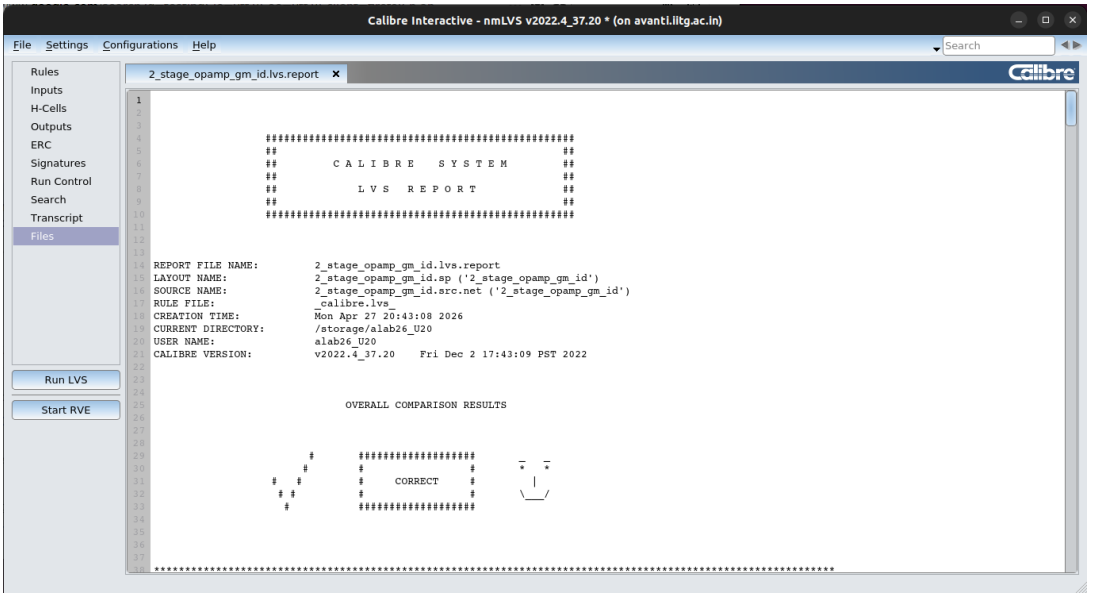


Figure 11: LVS

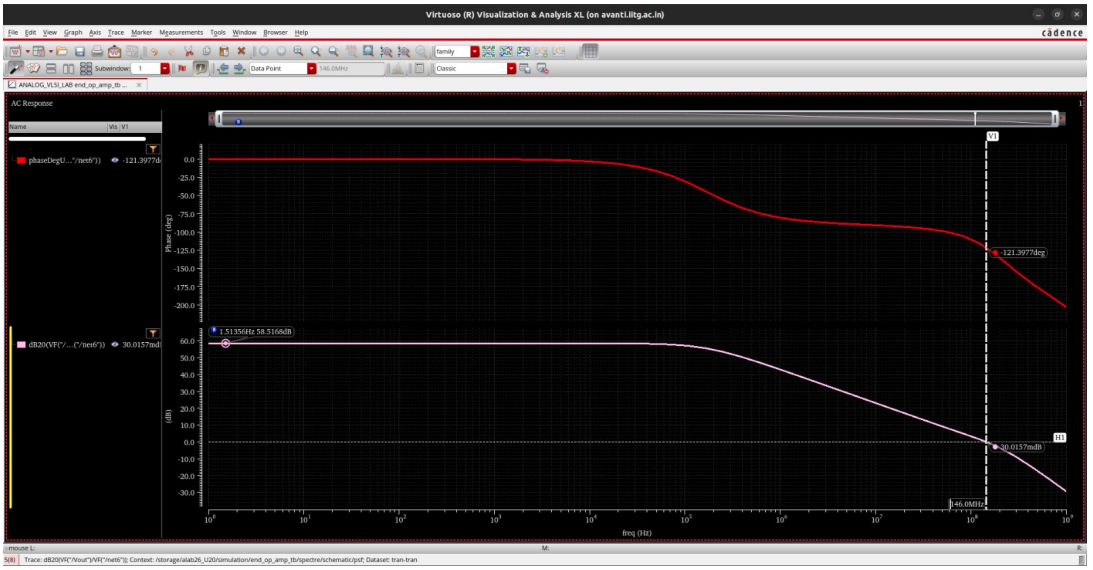


Figure 12: Post Pex gain phase GBW

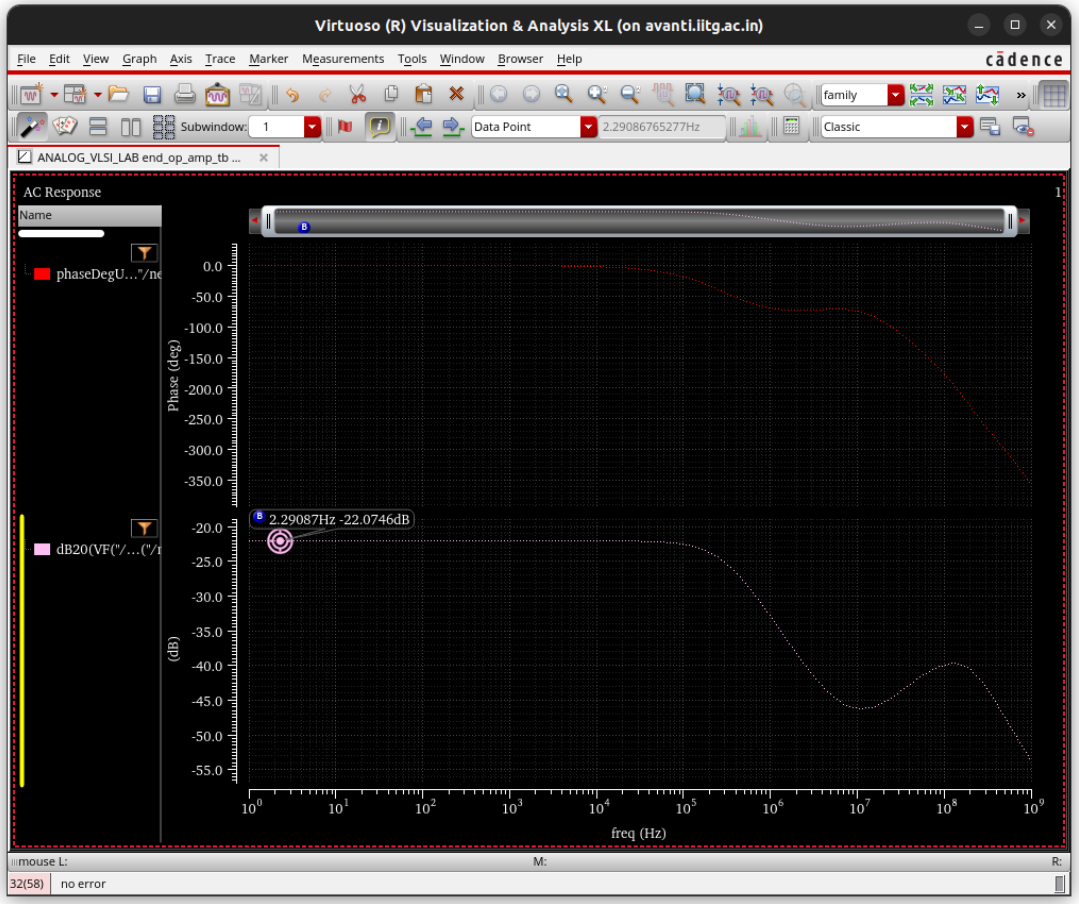


Figure 13: Post Pex CM gain

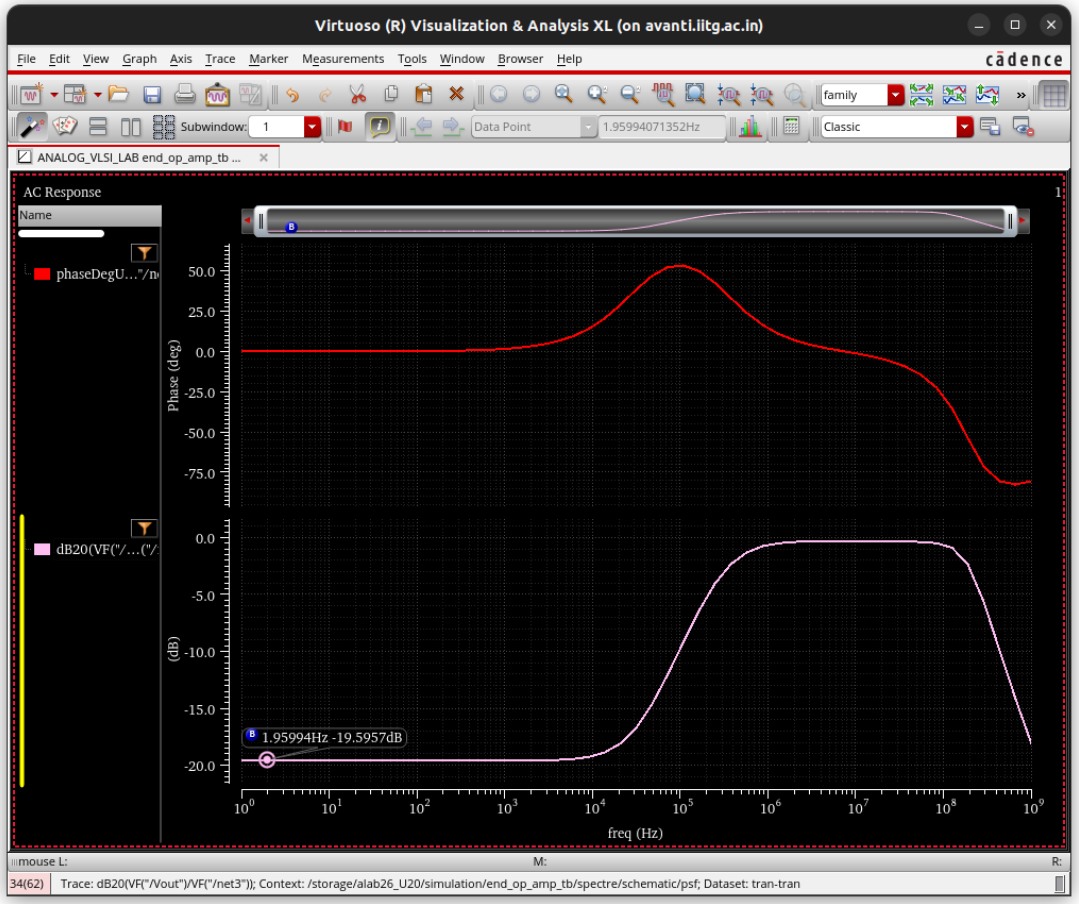


Figure 14: Post Pex PSRR

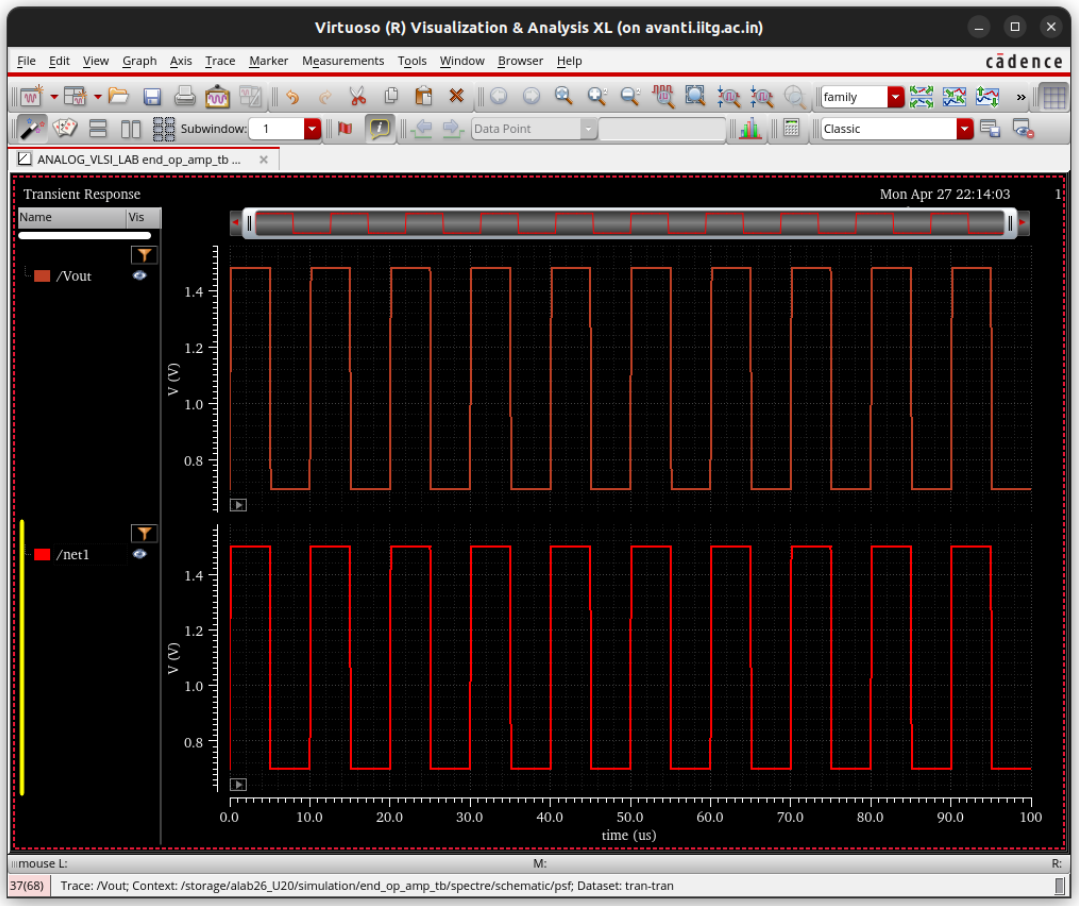


Figure 15: Post pex Pulse input(100khz) and Output

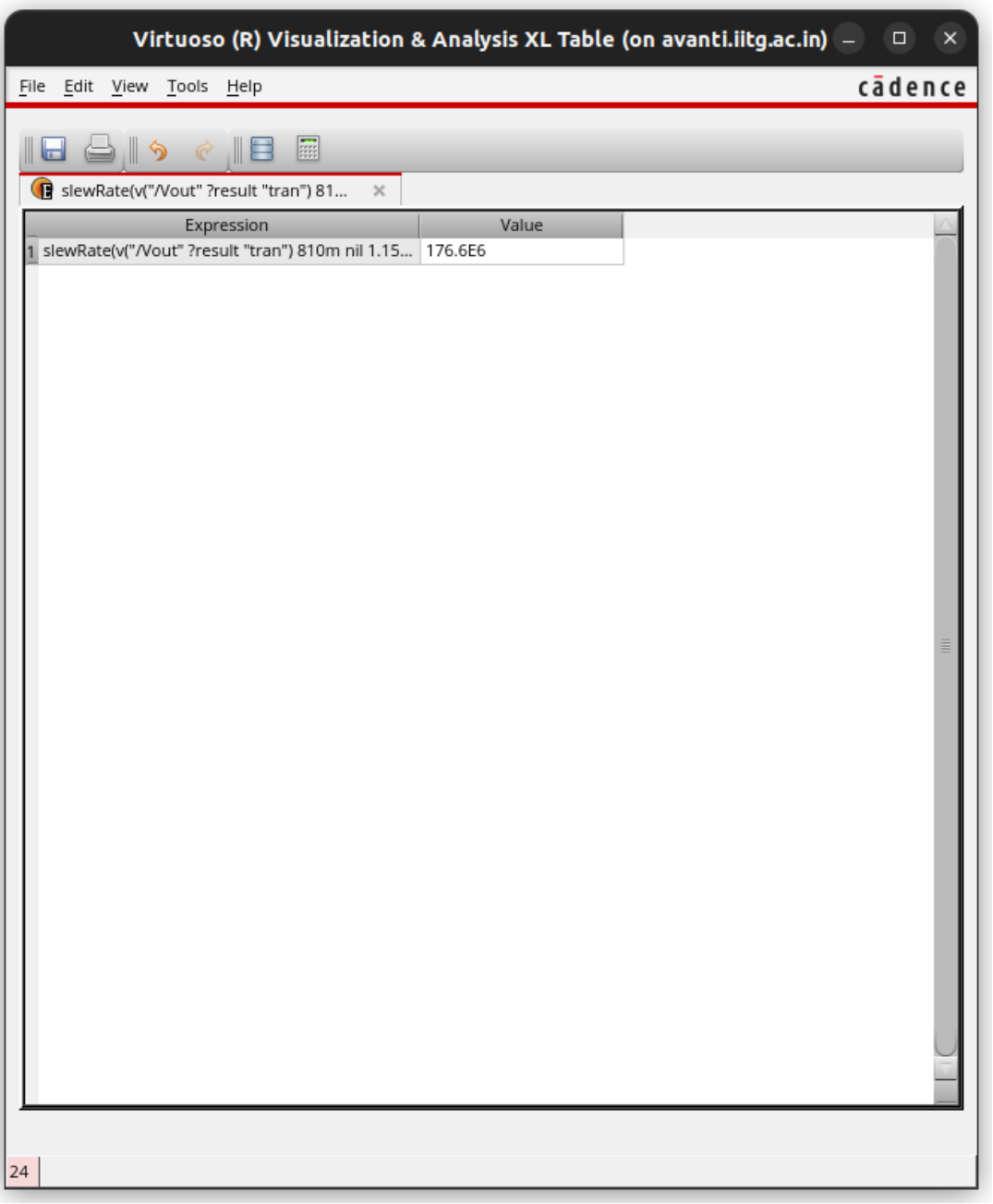


Figure 16: Post pex Slew Rate

5 Performance Comparison of 2 stage op amp pre vs post pex

Parameter	Pre Layout	Post Layout	Variation
DC Gain	60.6303 dB	58.5168 dB	slight decrease
Unity Gain Frequency	167 MHz	146 MHz	reduced
Phase Margin	56.61249 deg	58.6023 deg	slightly increased
Slew Rate	176.6 V/us	148.998 V/us	reduced
Common Mode Gain	-11.1724 dB	-22.0746 dB	improved
CMRR	67.785 dB	80.6769 dB	improved

Table 1: Performance comparison without null resistor